

Rossby Wave Momentum Flux

$$\text{Let } \psi = \text{Re } A e^{i(kx + ly - \omega t)}$$

$$u = -\frac{\partial \psi}{\partial y}, \quad v = \frac{\partial \psi}{\partial x}$$

A is complex

$$u = -\text{Re } i l A e^{i l}$$

$$v = \text{Re } i k A e^{i l}$$

To get $\overline{uv}$ must take the real parts

First. Let $A = a + ib$

$$\psi = \text{Re } (a + ib) e^{i l}$$

$$\therefore \psi = a \cos(kx + ly - \omega t) - b \sin(kx + ly - \omega t)$$

$$\text{Now, } u = -\partial \psi / \partial y = +a l \sin(\quad) + b l \cos(\quad) \quad (1)$$

$$v = \partial \psi / \partial x = -a k \sin(\quad) - b k \cos(\quad). \quad (2)$$

$$\text{Now, we need } \overline{uv} = \frac{1}{L} \int_0^L u v dx \quad L \text{ is a wave length}$$

$$\frac{1}{L} \int_0^L \sin^2 kx dx = \frac{1}{L} \int_0^L \cos^2 kx dx = 1/2 \quad (3)$$

$$\frac{1}{L} \int_0^L \sin \cos dx = 0 \quad (4)$$

Using (3), (4), in (1), (2), we obtain

$$\overline{u_v} = -\frac{1}{2} a' k l - \frac{1}{2} b' k l$$

$$= -\frac{1}{2} (a' + b') k l$$

$$\boxed{\overline{u_v} = -\frac{1}{2} A' k l}$$

Alternate

$$\text{let } \psi = B e^{i(kx + ly - \omega t)} + B^* e^{-i(kx + ly - \omega t)}$$

B^* = complex conjugate of B .

This is the same as before if $B = \frac{1}{2} A$.

$$u = -\frac{\partial \psi}{\partial y} = -B i l e^{i(\dots)} + i l B^* e^{-i(\dots)}$$

$$v = \frac{\partial \psi}{\partial x} = B i k e^{i(\dots)} - i k B^* e^{-i(\dots)}$$

Both expressions above are real

$$u v = \left[-B i l e^{i(\dots)} + i l B^* e^{-i(\dots)} \right] \times \left[B i k e^{i(\dots)} - i k B^* e^{-i(\dots)} \right]$$

$$u v = -2 B B^* k l + B^2 k l e^{2i(\dots)} + B^{*2} k l e^{2i(\dots)}$$

When we take the average only the first term contributes

$$\therefore \frac{1}{L} \int u v dx = -2 |B|^2 k l = -\frac{1}{2} |A|^2 k l$$

as before